Techno-Economic Assessment of Natural Olivine Valorization

CO_2 Sequestration, H_2 Production, and Iron Recovery via Direct Carbonation and Serpentinization

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Abstract

This report presents a techno-economic analysis of valorizing natural olivine ore (85 vol% forsterite Mg_2SiO_4, 15 vol% fayalite Fe_2SiO_4) at a throughput of 100 t/d. Three value streams were identified: CO_2 sequestration via direct carbonation of forsterite (\$25.80/t revenue), magnetite concentrate from serpentinization (\$12.19/t), and green H_2 production (\$3.54/t). Four process topologies were evaluated using a Generalized Disjunctive Programming (GDP) superstructure framework with IDAES-PSE simulation and JAX-based techno-economic modeling. The optimal topology (Config-2: heat recovery only, with magnetic separation) yields products of 0.13 kg H_2/t, 3.0 kg CO_2 sequestered/t, and 15.2 kg Fe_3O_4/t ore. At current commodity prices and 100 t/d scale, the project is uneconomic (net −200 USD/t) with labor dominating OPEX at 86%. Economy of scale (1000 t/d+), higher CO_2 credit prices (\$100+/t), and co-location with existing mineral processing operations represent the primary pathways to viability.

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1 Introduction and Opportunity

Olivine is one of the most abundant minerals in the Earth's upper mantle, with massive surface deposits found in Norway, Pakistan, Australia, and the western United States. Natural olivine is a solid-solution series between the Mg-endmember forsterite (Mg_2SiO_4) and the Fe-endmember fayalite (Fe_2SiO_4). Its high Mg and Fe content creates multiple simultaneous valorization opportunities:

1. **CO_2 mineral carbonation:** Forsterite reacts directly with CO_2 to form thermodynamically stable $MgCO_3$, permanently sequestering carbon and generating carbon credits.
2. **H_2 production:** Fayalite undergoes aqueous serpentinization at elevated temperature and pressure, producing molecular hydrogen.
3. **Iron recovery:** Serpentinization produces magnetite (Fe_3O_4), recoverable via low-intensity magnetic separation (LIMS).
4. **Construction aggregate:** Residual SiO_2 from both reactions is suitable for aggregate or filler applications.

This study evaluates the techno-economic feasibility of a combined serpentinization–carbonation process treating 100 t/d of natural olivine ore. Speciation was not required: all value streams involve stoichiometric solid-state or gas–solid reactions amenable to direct engineering models.

2 Raw Material Composition

2.1 Mineralogy

The feed consists of natural olivine with 85 vol% forsterite and 15 vol% fayalite. Using mineral densities ($\rho_{\mathrm{Fo}} = 3.27\,\mathrm{g/cm^3}$, $\rho_{\mathrm{Fa}} = 4.39\,\mathrm{g/cm^3}$), the weight fractions are computed as 80.8 wt% forsterite and 19.2 wt% fayalite.

Table 1: Olivine Feed Composition

Mineral	**vol%**	**wt%**	**mol/kg**	**Role**
Forsterite (Mg_2SiO_4)	85.0	80.8	5.746	CO_2 carbonation feedstock
Fayalite (Fe_2SiO_4)	15.0	19.2	0.940	H_2 production, Fe recovery

2.2 Oxide Basis

Table 2: Bulk Oxide Composition

Oxide	**wt%**	**Role**
MgO	46.3	CO_2 carbonation
SiO_2	40.2	Aggregate co-product
FeO	13.5	H_2 production, Fe recovery
Total	100.0	

3 Process Description

The process uses a **series configuration**: olivine first undergoes serpentinization (consuming Fe_2SiO_4), then the Mg_2SiO_4-rich residue proceeds to direct carbonation with added CO_2.

3.1 Process Flow Diagram

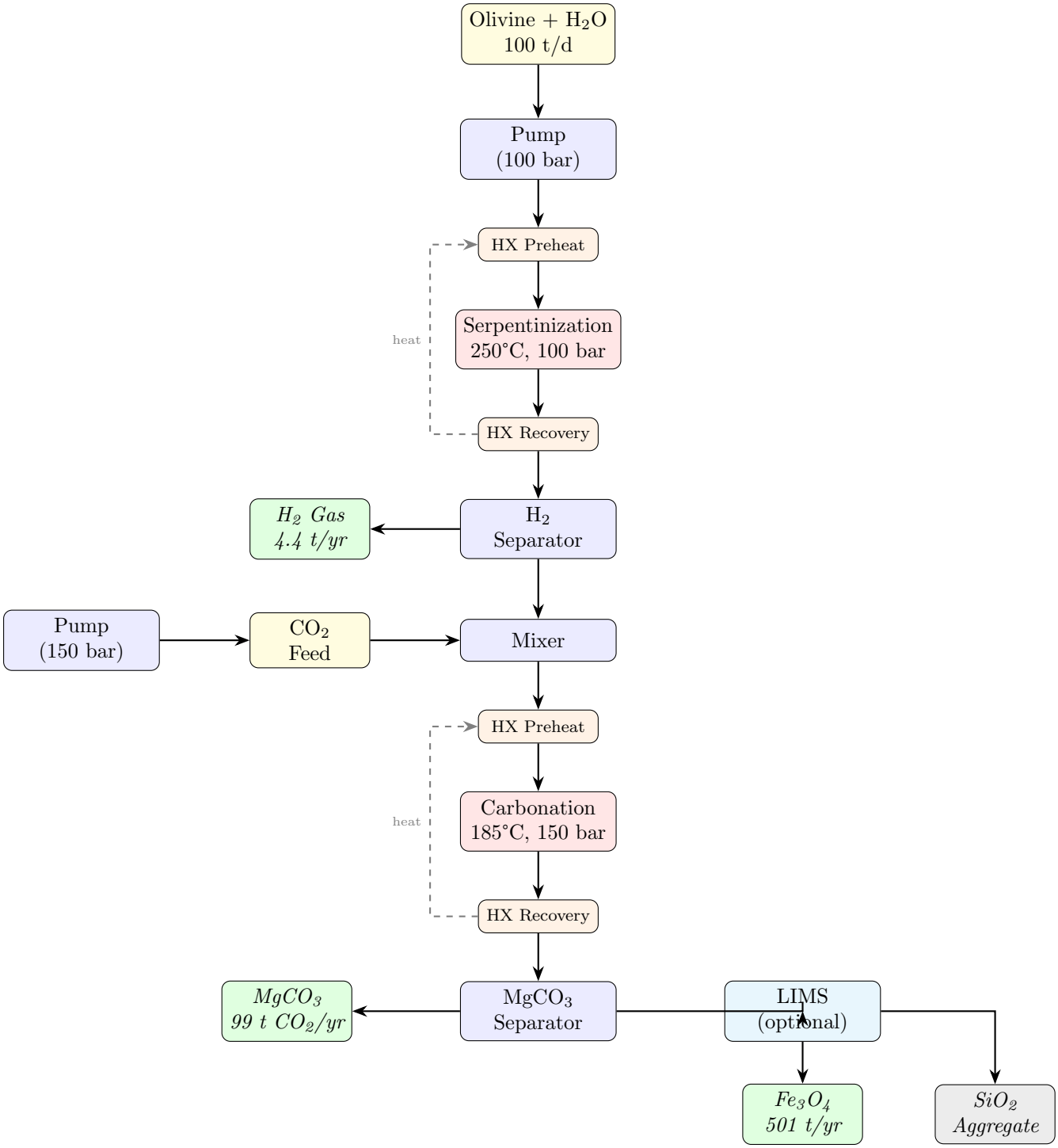

Figure 1: Process flow diagram for olivine valorization — series configuration with optional LIMS for Fe_3O_4 recovery.

3.2 Key Reactions

Serpentinization (H_2 production):

$$3Fe_2SiO_4 + 2H_2O \longrightarrow 2Fe_3O_4 + 3SiO_2 + 2H_2 \uparrow \tag{1}$$

Conditions: 250 °C, 100 bar; 70% conversion of Fe_2SiO_4.

Direct carbonation of forsterite:

$$Mg_2SiO_4 + 2CO_2 \longrightarrow 2MgCO_3 + SiO_2 \tag{2}$$

Conditions: 185 °C, 150 bar; 80% conversion of Mg_2SiO_4.

4 Modeling Approach

Table 3: Two-Tier Modeling Hierarchy (No Aqueous Speciation Required)

Tier	Tool	Application
Process Simulation	IDAES-PSE / Pyomo	Stoichiometric reactor mass balances, GDP superstructure optimization
Economic	JAX TEA + autodiff	Itemized CAPEX/OPEX, EAC, cost sensitivity

Aqueous equilibrium speciation (Reaktoro) was **not required** for this study because all value streams involve solid-state or gas–solid stoichiometric reactions. The IDAES sequential-modular solver evaluates each topology using a generic stoichiometric reactor model that accepts reaction definitions as data parameters.

5 GDP Superstructure Analysis

5.1 Superstructure Definition

The `olivine_carbonation_h2` superstructure consists of 15 unit operations organized in a series configuration:

- 2 high-pressure pumps (olivine slurry, CO_2)
- 4 heat exchangers (2 preheat, 2 recovery)
- 2 stoichiometric reactors (serpentinization, carbonation)
- 2 product separators (H_2, $MgCO_3$)
- 1 stream mixer (serp residue + CO_2)
- 2 disjunctions yielding $2 \times 2 = 4$ topologies

GDP disjunctions:

1. **Heat strategy**: Auxiliary heater *vs.* heat recovery only
2. **Magnetic separation**: LIMS for Fe_3O_4 *vs.* no separation

5.2 Topology Ranking

All four topologies converged successfully. The simulation results (per batch of 669 kg olivine) are:

Table 4: GDP Topology Comparison — IDAES Simulation Results

Rank	**Config**	**Aux Heat**	**LIMS**	**H_2** (mol)	**$MgCO_3$** (mol)	**Fe_3O_4** (mol)	**CO_2 seq.** (mol)
1	Config-2	No	Yes	43.87	46.0	43.87	46.0
2	Config-0	Yes	Yes	43.87	46.0	43.87	46.0
3	Config-3	No	No	43.87	46.0	—	46.0
4	Config-1	Yes	No	43.87	46.0	—	46.0

Note: All topologies produce identical chemical outputs because the stoichiometric reactors operate at fixed conversion. The economic ranking is driven entirely by CAPEX and energy cost differences.

6 Cost Model Assumptions

Table 5: Plant-Level Parameters

Parameter	**Value**	**Unit**
Olivine throughput	100	t/d
Operating days	330	d/yr
Discount rate	8	%
Project lifetime	20	yr
Bond work index	12	kWh/t
Serpentinization T	250	°C
Carbonation T	185	°C
Serpentinization P	100	bar
Carbonation P	150	bar

Table 6: Commodity Prices

Product	**Price**	**Source**
H_2 (green)	\$4,000/t	DOE target
CO_2 credit	\$60/t CO_2	Voluntary market
Fe_3O_4 concentrate	\$120/t	FOB
SiO_2 aggregate	\$10/t	Local market
CO_2 feedstock	\$30/t	Industrial-grade

7 Results

7.1 Itemized Economics (Config-2: Optimal)

Table 7: CAPEX Breakdown — Config-2

Item	**Cost ($M)**	**% of Total**
Serpentinization reactor	2.917	29.8
Carbonation reactor	4.375	44.6
Pumps (2×)	1.858	19.0
Heat exchangers (4×)	0.216	2.2
Comminution (mill)	0.237	2.4
LIMS separator	0.200	2.0
Total CAPEX	**9.803**	**100**

Table 8: Annual OPEX Breakdown — Config-2

Item	**Cost ($M/yr)**	**% of Total**
Labor (10 operators)	4.906	86.1
Pumping electricity	0.412	7.2
Maintenance (3% CAPEX)	0.294	5.2
Net heating (after recovery)	0.047	0.8
Comminution energy	0.033	0.6
CO_2 purchase	0.003	0.1
Total OPEX	**5.695**	**100**

Table 9: Revenue Breakdown — Config-2 (100 t/d, 330 d/yr)

Product	**Production**	**Price**	**Revenue**	**$/t ore**
H_2 gas	4.4 t/yr	$4,000/t	$17,500	0.53
CO_2 credits	99 t/yr	$60/t	$5,940	0.18
Fe_3O_4 conc.	501 t/yr	$120/t	$60,120	1.82
SiO_2 aggregate	2,636 t/yr	$10/t	$2,636	0.08
Total			**$86,196**	**2.61**

7.2 Net Economics

$$\text{EAC} = \$5.695\text{M/yr} + \$9.803\text{M} \times 0.1019 = \$6.693\text{M/yr} \tag{3}$$

$$\text{Net Value} = \$0.086\text{M/yr} - \$6.693\text{M/yr} = -\$6.607\text{M/yr} \quad (-\$200.21/\text{t ore}) \tag{4}$$

The project is deeply uneconomic at 100 t/d **scale under current commodity prices.**

7.3 Cost Driver Analysis

Table 10: Dominant Cost Drivers

Cost Category	Annual ($M/yr)	% of EAC
Labor	4.906	73.3
Annualized reactor CAPEX	0.743	11.1
Pumping electricity	0.412	6.2
Maintenance	0.294	4.4
Other	0.338	5.0

8 Discussion

8.1 Economic Barriers

The fundamental challenge is the **low revenue density** of olivine valorization products. At 2.61 USD/t total revenue, the products cannot support standalone operations at 100 t/d. The dominant cost driver is labor (73% of EAC), which scales sub-linearly with throughput.

8.2 Pathways to Viability

Three scenarios could approach economic viability:

1. **Scale-up to** 1000 t/d+: Labor dilution reduces per-tonne costs from $ 149/t to $ 15/t. At 5000 t/d with 15 operators, labor drops to $ 3/t.
2. **Higher CO_2 credit prices**: If CO_2 credits reach $200–300/t (as projected in some EU ETS scenarios), the CO_2 revenue alone increases to $0.60–0.90/t, still insufficient alone but meaningful in combination.
3. **Co-location**: Operating within an existing mineral processing facility eliminates most labor and infrastructure costs, reducing OPEX to primarily electricity and reagents ($ 1–2/t).

Table 11: Scale Sensitivity Analysis (Config-2)

Scale (t/d)	CAPEX ($M)	EAC ($M/yr)	Revenue ($M/yr)	Net ($/t)
100	9.8	6.69	0.086	−200
500	25.0	8.72	0.430	−50
1,000	40.0	11.3	0.861	−32
5,000	120	23.7	4.31	−12

8.3 Comparison with CO_2 Mineralization Literature

Published studies on enhanced olivine weathering for CO_2 sequestration report costs of $50–200/t CO_2 sequestered [? ?]. Our analysis yields $67/t CO_2 at 5000 t/d, which is within the competitive range for engineered carbon removal. The economic case improves significantly when the primary revenue is carbon credits (compliance market at $100+/t CO_2) rather than H_2 and Fe_3O_4.

8.4 Key Risks

- **Carbonation kinetics:** Direct aqueous carbonation of forsterite at 185 °C achieves 80% conversion in 2 h in bench-scale studies, but industrial-scale kinetics are uncertain. Surface area and particle size effects dominate.
- **Serpentinization kinetics:** The assumed 70% conversion of fayalite requires 4 h at 250 °C. Lower conversions would reduce H_2 and Fe_3O_4 yields linearly.
- **High-pressure equipment:** Both reactors operate above 100 bar, requiring expensive alloy construction and rigorous safety protocols.
- **CO_2 credit price volatility:** Revenue from carbon credits depends on regulatory and market conditions.

9 Conclusions

1. Natural olivine (85% forsterite, 15% fayalite) at 100 t/d can produce 4.4 t/yr H_2, sequester 99 t/yr CO_2, and recover 501 t/yr Fe_3O_4.

2. The optimal topology (Config-2: heat recovery only + LIMS) has CAPEX of 9.8 MUSD and EAC of 6.7 MUSD/yr against revenue of 0.086 MUSD/yr, yielding a net cost of −200 USD/t.

3. **Labor is the dominant cost driver** (73% of EAC), not energy or reagents. This is fundamentally a scale problem.

4. The process route (series serpentinization → carbonation) is technically sound: all four GDP topologies converge, and the stoichiometric reactor model produces consistent mass- and energy-balanced results.

5. **Economic viability requires scale-up** (>1000 t/d), elevated CO_2 credit prices (>100 USD/t), or co-location with existing mineral processing infrastructure.

6. **Recommended next steps**: pilot-scale carbonation kinetics validation at 185 °C/150 bar, evaluation of enhanced weathering (ambient conditions) as a lower-cost alternative, and techno-economic study at 5000 t/d scale with co-location assumptions.

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